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DEPARTMENT OF MATHEMATICS AND APPLIED MATHEMATICS

Mathematics II

Advanced Calculus (2AC)

Change of variables in double integrals

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0.1 Introduction

Sometimes the domain D of integration for a double integral can be very complicated for the evaluation of the double integral $\iint_D f(x, y) dx dy$. The purpose of the change of variables is to find a suitable transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ with $(x, y) = T(s, t)$ such that $D = T(R)$ that is, the domain D in the xy -plane is the image of some domain R in the st -plane. Then, we write the double integral as a double integral over the new domain R , of a new function of the variables s and t .

We studied in the first year calculus course for the case of the integral of a function of one variable that

$$\int_a^b g(x) dx = \int_{h^{-1}(a)}^{h^{-1}(b)} g(h(t)) h'(t) dt, \quad x = h(t) \quad (1)$$

for a differentiable monotone (strictly increasing or decreasing) $h : I \rightarrow [a, b]$ with $I = [h^{-1}(a), h^{-1}(b)]$ or $I = [h^{-1}(b), h^{-1}(a)]$ according to whether h is strictly increasing or decreasing.

Following a similar procedure, we write

$$(x, y) = T(s, t) = (x(s, t), y(s, t))$$

so that we get the expression $f(x, y) = f(x(s, t), y(s, t))$.

The question now is: what will be the equivalent of $h'(t)$ in (1) for the double integral?

The answer to that question is found in the following theorem.

Theorem 0.1 (Change of variables in double integrals)

Let $f : D \subset \mathbb{R}^2 \rightarrow \mathbb{R}$ be a continuous function. Let $T : R \subset \mathbb{R}^2 \rightarrow D$ be a function with $(x, y) = T(s, t)$. Assume that

- (i) T is invertible;
- (ii) T and T^{-1} are of class $\mathcal{C}^1$ i.e., both functions are differentiable with continuous first order partial derivatives;

$$(iii) \quad J_T(s, t) := \det \left(\frac{\partial(x, y)}{\partial(s, t)}(s, t) \right) := \begin{vmatrix} x_s(s, t) & x_t(s, t) \\ y_s(s, t) & y_t(s, t) \end{vmatrix} \neq 0 \quad \left\{ \begin{array}{l} \text{except on a finite number} \\ \text{of lines in the domain } R. \end{array} \right.$$

Then,

$$\boxed{\iint_D f(x, y) dx dy = \iint_R f(x(s, t), y(s, t)) |J_T(s, t)| ds dt} \quad (2)$$

J_T is called the **Jacobian** of the transformation T .

Remark 0.2 As we have seen, a double integral is generated as limit of double Riemann sums. So, if we consider a partition of R into subrectangles $(R_{ij})_{1 \leq i \leq n, 1 \leq j \leq m}$ to which there corresponds a partition of the domain D through the transformation T into patches $(D_{ij})_{1 \leq i \leq n, 1 \leq j \leq m}$.

The magnitude of the Jacobian of T represents at the level of the Riemann sums, the scalar factor between the area of each subrectangle R_{ij} in the partition of the rectangle R and the area of the corresponding patch D_{ij} in D through the transformation T . That is,

$$A(D_{ij}) \approx |J_T(s_i^*, t_j^*)| A(R_{ij})$$

and hence,

$$\sum_{i=1}^n \sum_{j=1}^m f(x_i^*, y_j^*) A(D_{ij}) = \sum_{i=1}^n \sum_{j=1}^m f(T(s_i^*, t_j^*)) |J_T(s_i^*, t_j^*)| A(R_{ij}), \quad (3)$$

where (x_i^*, y_j^*) is sample point in the subrectangle R_{ij} .

Now, passing to the limit in (3), we obtain the change of variables identity (2).

Example 0.3 Evaluate the double integral $\iint_D y x dx dy$, where D is the domain enclosed by the curves $y = \frac{1}{x}$, $y = \frac{2}{x}$, $y = \frac{1}{x^2}$ and $y = \frac{2}{x^2}$.

Solution: We write D as

$$D = \{(x, y) \in \mathbb{R}^2: 1 \leq xy \leq 2, 1 \leq x^2 y \leq 2\}.$$

It is natural to consider here the change of variables

$$\begin{aligned} T^{-1}: \begin{cases} s := xy \\ t := x^2 y \end{cases} &\Rightarrow T: \begin{cases} x = \frac{t}{s}, \\ y = \frac{s^2}{t} \end{cases} \Rightarrow D = T(R) \quad \text{with} \\ R &:= \{(s, t) \in \mathbb{R}^2: 1 \leq s \leq 2, 1 \leq t \leq 2\} \\ \Rightarrow J_T(s, t) &= \begin{vmatrix} x_s(s, t) & x_t(s, t) \\ y_s(s, t) & y_t(s, t) \end{vmatrix} = \begin{vmatrix} -\frac{t}{s^2} & \frac{1}{s} \\ \frac{2s}{t} & -\frac{s^2}{t^2} \end{vmatrix} = \frac{ts^2}{s^2 t^2} - \frac{2s}{st} = \frac{1}{t} - \frac{2}{t} = -\frac{1}{t}. \end{aligned}$$

Therefore,

$$\begin{aligned} \iint_D y x dx dy &= \iint_R s |J_T(s, t)| ds dt = \int_1^2 s \left(\int_1^2 \frac{1}{t} dt \right) ds = \int_1^2 s [\ln t]_1^2 ds \\ &= \int_1^2 s \ln 2 ds = \left[\frac{1}{2} s^2 \right]_1^2 \ln 2 = \left(2 - \frac{1}{2} \right) \ln 2 = \frac{3}{2} \ln 2. \end{aligned}$$

0.2 Evaluation of a double integral through the use of polar coordinates

An important application of the change of variables in double integrals is the evaluation of a double integral through polar coordinates. This happens when the boundary of the domain D involves for instance an arc of a circle.

Example 0.4 Let us evaluate the double integral $\iint_D (y^2 + 1) dx dy$ where

$$D := \left\{ (x, y) \in \mathbb{R}^2 : x^2 + y^2 \leq 4, -x \leq y \leq x \right\}.$$

Solution: Using polar coordinates $T : \begin{cases} x = r \cos \theta, \\ y = r \sin \theta, \end{cases}$ we find that $D = T(R)$ where R is the rectangle

$$R := \left\{ (r, \theta) \in \mathbb{R}^2 : 0 \leq r \leq 2, -\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4} \right\}.$$

We find that the mapping T has Jacobian

$$J_T(r, \theta) = \begin{vmatrix} x_r(r, \theta) & x_\theta(r, \theta) \\ y_r(r, \theta) & y_\theta(r, \theta) \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r.$$

So,

$$\begin{aligned} \iint_D (y^2 + 1) dx dy &= \iint_R (r^2 \sin^2 \theta + 1) r dr d\theta = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \int_0^2 [r^3 \sin^2 \theta + r] dr d\theta \\ &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left[\frac{1}{4} r^4 \sin^2 \theta + \frac{1}{2} r^2 \right]_0^2 d\theta = \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} [4 \sin^2 \theta + 2] d\theta = 2 \int_0^{\frac{\pi}{4}} [4 \sin^2 \theta + 2] d\theta \\ &= 2 \int_0^{\frac{\pi}{4}} [4 - 2 \cos(2\theta)] d\theta = 2 \left[4\theta - \sin(2\theta) \right]_0^{\frac{\pi}{4}} = 2(\pi - 1) = 2\pi - 2. \end{aligned}$$

Exercises 0.5

1. Evaluate the double integral $\iint_D f(x, y) dx dy$ where:

(a) $f(x, y) := x - y$ and $D := \{(x, y) \in \mathbb{R}^2 : -1 \leq 2x + y \leq 1, 0 \leq -x + y \leq 2\}$;

Answer: $\boxed{-\frac{4}{3}}$ using a suitable change of variable.

(b) $f(x, y) = y^2 + 1$ and $D := \{(x, y) \in \mathbb{R}^2 : x^2 + 4y^2 \leq 1, x \leq 0, y \geq 0\}$.

Answer: $\boxed{\frac{17}{128}\pi}$.

(c) $f(x, y) = 2y\sqrt{x}$ and $D := \{(x, y) \in \mathbb{R}^2 : x^2 + y^2 \geq 1, 0 \leq x \leq 1, 0 \leq y \leq 1\}$.

Answer: $\boxed{\frac{2}{7}}$

(d) $f(x, y) = e^{y^2}$ and $D := \{(x, y) \in \mathbb{R}^2 : |x| \leq y \leq 1\}$. Answer: $\boxed{e - 1}$

(e) $f(x, y) := x\sqrt{x^2 + y^2}$ and $D := \{(x, y) \in \mathbb{R}^2: x^2 + y^2 \leq 1, x^2 + y^2 \leq 2y, x \leq 0\}$.

Answer: $\boxed{-\frac{3}{20}}$.

2. Find the area of the domain D in the xy -plane enclosed by the curves $xy = a^2$ and $xy = b^2$ with $(0 < a < b)$ and the lines $y = \alpha x$ and $y = \beta x$ with $(0 < \alpha < \beta)$.

Hint: $A(D) := \iint_D 1 dx dy$ and use a suitable change of variables. Answer: $\boxed{\frac{b^2 - a^2}{2} \ln\left(\frac{\beta}{\alpha}\right)}$.

The next set of notes will be on triple integrals